Algorithm 1: $G_k \leftarrow$ `ChooseSequentialRootWidth`$(\texttt{WidthState}_{k-1})$

Input: normal or sequential-RB mode; configured integers $1 \le G_{\min} \le G_{\max}$; device set $\mathcal{G}$; checkpointed controller state from completed iterations only

Hyperparameters: per-device capacity $M_{\mathrm{device},g}$, unused guard $M_{\mathrm{guard},g}$, growth fraction $\rho \in (0,1)$, successful-iteration growth delay h, one-sided prediction level z, and calibration window W

Definitions: G is the common number of new root trajectories per GPU for one iteration, so the distributed iteration contains $|\mathcal{G}|G$ roots. The hard ceiling is $M_{\mathrm{hard},g} = M_{\mathrm{device},g} - M_{\mathrm{guard},g}$

Definitions: For phase $\phi \in \{\mathsf{rollout}, \mathsf{seal}, \mathsf{targets}, \mathsf{optimization}\}$, $\widehat{M}^{+}_{g,\phi,k}(G)$ is a monotone upper prediction formed from the exact byte formulas and reserved capacities of `storage_lifecycle.tex`. Variable counts per root use lagged one-sided estimates $\widehat{\mu} + z\widehat{s}$ for decisions, leaves, occurrences, summary-anchor bytes, chunk slack, seal destinations/scratch, posterior rows, and physical workspace. Fixed model/optimizer/collective bytes are measured directly. Positive residuals widen the bound immediately; a bound may tighten only after W successful observations

Definitions: The optional reference cache is excluded from this feasibility proof. Optimization must always admit the uncached reference path; a cache may be added later only by the hard transactional test in `DistributedFixedRolloutUpdate` from `distributed_update.tex`

Output: one width fixed before any current-iteration random draw, or a clean pre-rollout stop if even $G_{\min}$ is unsafe

foreach $g \in \mathcal{G}$ ***in parallel*** **do**

  Refresh immutable schema bytes, allocator/backend calibration, and the current usable device ceiling; read no statistic from iteration k;

  if *no completed calibration is available* **then**

    Evaluate $G_{\min}$ with the declared analytic per-root worst-case geometry and conservative singleton workspaces; set $\widetilde{G}_g \leftarrow G_{\min}$ only if every phase fits, otherwise set it to zero; initialize $G_{k-1} \leftarrow G_{\min}$ and mark growth disabled;

  else

    By integer monotone search compute

$$\widetilde{G}_g \leftarrow \max\left\{ G \in [G_{\min}, G_{\max}] : \max_{\phi} \widehat{M}^{+}_{g,\phi,k}(G) \le M_{\mathrm{hard},g} \right\};$$

    if this set is empty, set $\widetilde{G}_g \leftarrow 0$;

  end

end

$G^{\mathrm{safe}} \leftarrow \texttt{AllReduceMin}_g\, \widetilde{G}_g$ `/* identical work width and collective order on every rank */`

if $G^{\mathrm{safe}} < G_{\min}$ **then**

  checkpoint and stop before roots, queues, or rollout storage are allocated

end

if $G^{\mathrm{safe}} < G_{k-1}$ **then**

  $G_k \leftarrow G^{\mathrm{safe}}$ `/* unsafe evidence causes an immediate decrease */`

else if *the preceding h completed iterations were below their predicted phase ceilings and growth is enabled* **then**

  $G_k \leftarrow \min\{G^{\mathrm{safe}}, G_{\max}, \max(G_{k-1}+1, \lfloor (1+\rho)G_{k-1} \rfloor)\}$;

else

  $G_k \leftarrow G_{k-1}$;

end

Freeze G_k in the iteration manifest and address roots by (g,i), $0 \le i < G_k$; no later admission decision may change this set; **return** G_k;

Algorithm 2: $\texttt{WidthState}_k \leftarrow$ `ObserveSequentialMemory`$(G_k, \texttt{IterationLedger}_k)$

Input: a fully successful synchronized iteration; exact logical counts, allocated capacities, allocator-reserved and NVML phase high-water marks, physical-plan variants, and backend workspace observations from every GPU

Output: checkpointable state used only for iteration $k+1$ and later

For each GPU and phase, validate the ledger against the immutable storage schema; missing, nonfinite, or inconsistent accounting is unsafe evidence; Update per-root count/byte estimators and one-sided residual envelopes. Apply upward surprises immediately; age out an upper observation only after at least W later successful observations preserve coverage; Record which phase and field constrained the next width, the predicted and observed peaks, guard headroom, controller version, and G_k; Checkpoint the state atomically with the training checkpoint; `/* Never discard, truncate, reroll, or retry an iteration to make its realized geometry fit. A runtime singleton/admission failure is a clean checkpointed stop and conservative next-run evidence, not a new random draw. */`

return $\mathit{WidthState}_k$;